Homework for Chap 9

In a piece of semiconductor of GaAs, given the static dielectric constant is 12.9, the effective mass for electron is $0.067m_0$, the effective masses for heavy hole and light hole are $0.45m_0$ and $0.082\ m_0$, respectively. Here m_0 is the rest mass of electron, and $m_0=9.11\text{x}10^{-31}$kg. Calculate the Bohr radius and binding energy of exciton ground state in GaAs. Please note that electron can form exciton with both heavy hole and light hole.

徐圣毅 12310833 $m_{lh} = 0.082 m_0$

According to parameters: $m_e = 0.067 m_0$ $m_{hh} = 0.45 m_0$ $\cancel{m_{lh} = 0.082 m_0}$

The reduced exciton mass:

$$\mu = \frac{m_e m_h}{m_e + m_h}$$

$$\mu_{elh} = \frac{0.067 m_0 \cdot 0.082 m_0}{(0.067 + 0.082) m_0} \approx 0.0369 m_0$$

$$\mu_{ehh} = \frac{0.45 m_0 \cdot 0.067 m_0}{(0.45 + 0.067) m_0} \approx 0.0583 m_0$$

According to the definition of binding energy:

$$E_B = E_g - E_{ex} = Ry \cdot \frac{1}{n_B^2} - \frac{\hbar^2 K^2}{2M}$$

For direct gap ground state ($K = 0$, $n_B = 1$):

$$E_B = Ry = 13.6\,eV \cdot \frac{\mu}{m_0} \cdot \frac{1}{\varepsilon^2}$$

So: $$E_B^{elh} = 13.6\,eV \cdot \frac{0.0369 m_0}{m_0} \cdot \frac{1}{(12.9)^2} \approx 0.003\,eV = 3\,meV$$

$$E_B^{ehh} = 13.6\,eV \cdot \frac{0.0589 m_0}{m_0} \cdot \frac{1}{(12.9)^2} \approx 0.0048\,eV = 4.8\,meV$$

For exciton Bohr Radius:

$$a_B^{ex} = a_B^H \cdot \varepsilon \frac{m_0}{\mu} \quad , \quad a_B^H = 5.29 \times 10^{-11}\,m$$

$\therefore$ $a_B^{elh} = 5.29 \times 10^{-11} \cdot 12.9 \cdot \frac{m_0}{0.0369 m_0} \approx 18.5\,nm$ $a_B^{ehh} = 5.29 \times 10^{-11} \cdot 12.9 \cdot \frac{m_0}{0.0589 m_0} \approx 11.7\,nm$